24. Final Review

April 12, 2024 9:03 AM

1. Short Answer Questions (20 pt)

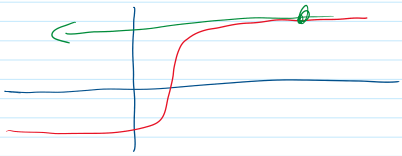
a. What is the difference between the Kiefer-Wolfowitz method and the Random Directions method? And what are the strengths of each?

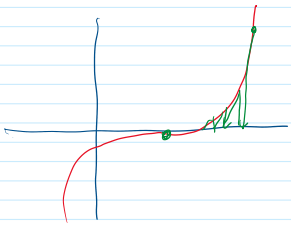
b. Give an example of a function were Newton-Raphson will likely fail to converge. Explain why.

- What works well with NR?

  Linear Function

- what works poorly?

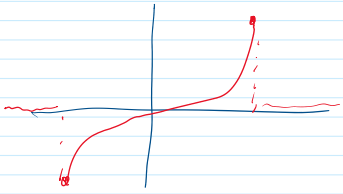




(Fails because $f' \sim 0$

so $\frac{1}{f'}$ gets really big.

Alternatively: NR doesn't respect boundaries



it can fall outside of the domain.

2. Ratio of Normals (20 pt)

Let $X, Y \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$. Show that

$$\left[Z = \frac{X-Y}{X+Y} \sim \text{Cauchy}. \right]$$

Hint: Try change of variables with $W = X + Y$, and recall that the normal pdf and Cauchy pdfs are[1]

$$f_{\text{normal}}(t) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{t^2}{2}\right) \text{ and } f_{\text{cauchy}}(t) = \frac{1}{\pi}\frac{1}{1+t^2}.$$

[1]Second Hint: $\int_{-\infty}^{\infty} \frac{|t|}{\sqrt{2\pi}} e^{-t^2/2} dt = \sqrt{\frac{2}{\pi}}$

$Z = \frac{X-Y}{X+Y=W}$

Step 1: get X, Y as functions of W, Z

$W = X + Y$

$$\begin{aligned} w &= X + Y \\ + \quad zw &= X - Y \\ \hline \frac{w(1+z)}{2} &= \frac{\cancel{2}X}{\cancel{2}} \end{aligned} \qquad \begin{aligned} w &= X + Y \\ - \quad zw &= X - Y \\ \hline \frac{w(1-z)}{2} &= \frac{\cancel{2}Y}{\cancel{2}} \end{aligned}$$

$$f_{wz}(w,z) = f_{XY}(X(w,z), Y(w,z))\,|J|$$

$$J = \begin{pmatrix} \frac{1+z}{2} & w/2 \\ \frac{1-z}{2} & -w/2 \end{pmatrix}, \quad |J| = \frac{|w|}{2}$$

$$e^{-y^2/2}$$

$$f_{wz}(w,z) = \frac{1}{\sqrt{2\pi}} e^{-w^2(1+z)^2/8} \frac{1}{\sqrt{2\pi}} e^{-w^2(1-z)^2/8} \frac{|w|}{2}$$

$$= \frac{1}{2\pi} \exp\left(-\frac{w^2}{4}(1+z^2)\right) \frac{|w|}{2}$$

How to get just $f_z(z)$?

$$f_z(z) = \frac{1}{2} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} |w| \exp\left(-\frac{w^2}{2}\left(\frac{1+z^2}{2}\right)\right) dw$$

- Do change of variables
- Use Hint to get $f_z(z) = \frac{1}{\pi} \frac{1}{1+z^2}$

3. Importance Sampling (20 pt)

We want to estimate the integral[2]

$$\int_0^1 e^x \sin(\pi x)\,dx.$$

Which of the following sampling distributions on $[0,1]$ results in the smallest variance?[3]

$$f_1(x) = \frac{2}{e^2 - 1} e^{2x} \quad \text{or} \quad f_2(x) = 2\sin^2(\pi x)$$

Variance $\sim \int_0^1 \frac{g(x)^2}{f(x)^2} f(x)\,dx = \int_0^1 \frac{g^2(x)}{f(x)}\,dx$

(1) $$\int_0^1 \frac{\cancel{e^{2x}}\sin^2(\pi x)}{\frac{2}{e^2-1}\,\cancel{e^{2x}}}\,dx = \frac{e^2-1}{2}\int_0^1 \frac{1-\cos(2\pi x)}{2}\,dx$$

$$= \frac{e^2-1}{2}\left(\underbrace{\frac{x}{2}}_{1/2} - \underbrace{\frac{\sin(2\pi x)}{4\pi}}_{0}\right)\Bigg|_0^1$$

$$= \frac{e^2-1}{4}$$

Do for f_2 and compare answers.

4. Statification (20 pt)

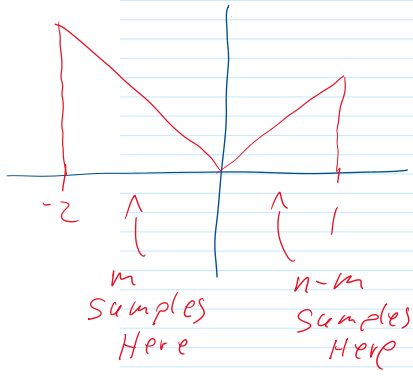
We want to use Monte Carlo to estimate the integral of

$$f(x) = \begin{cases} -x & x \in [-2,0] \\ x & x \in [0,1] \end{cases}$$

on the interval $[-2,1]$ using the stratified estimator

$$\hat{\mu} = \frac{2}{m}\sum_{i=1}^{m}(-X_i) + \frac{1}{n-m}\sum_{j=1}^{n-m} Y_j$$

where n is fixed, $m \in [1, n-1]$, $X_i \overset{\text{iid}}{\sim}$ Uniform $[-2,0]$, and $Y_i \overset{\text{iid}}{\sim}$ Uniform $[0,1]$. Find the value of m that minimizes the variance. That is $m = \delta n$ for some $\delta \in (0,1)$, which may not be an integer.



Note: $U \sim \text{Unif}[a,b]$, $\quad Var(U) = \frac{(b-a)^2}{12}$

- Compute $Var(\hat{\mu})$

$$= \frac{4}{m^2}\sum_{i=1}^{m}\underbrace{Var(X_i)}_{\frac{4}{12}} + \frac{1}{(n-m)^2}\sum_{j=1}^{n-m}\underbrace{Var(Y_j)}_{\frac{1}{12}}$$

- Find $m \in [1,n]$ that minimizes

$$\phi(m) = \frac{4}{3}\frac{1}{m} + \frac{1}{12}\frac{1}{(n-m)}$$

(If I recall, $m = \frac{4}{5}n$?)

5. At a loss (20 pt)

Do Calculus!!!

Consider the function, $g(t) = \sqrt{1+t^2} - 1$. Show that $g(t)$ is convex with minimum at $g(0) = 0$.

Then for data (x_i, y_i), find the gradient of

$$L(\theta) = \sum^{n}\left[\sqrt{1+\left(y_i - \sum^{k}\theta_i x_{i,i}\right)^2} - 1\right]$$

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$$L(\theta) = \sum_{i=1}^{n} \left[\sqrt{1 + \left(y_i - \sum_{j=1}^{k} \theta_j x_{i,j} \right)^2} - 1 \right]$$

$$\nabla_\theta L(\theta) = \nabla_\theta \left[\sum_{i=1}^{n} g\left(y_i - \sum_{j=1}^{n} \theta_j x_{ij} \right) \right]$$

$$= \sum_{i=1}^{n} \left[g'(-) \begin{pmatrix} | \\ -x_{ij} \\ | \end{pmatrix} \right]$$

jth entry

6. More Acceptance-Rejection (20 pt)

Let $f(t) = e^{-t}/(1+e^{-t})^2$ be the logistic distribution and $h(t) = e^{-|t|}/2$ be the Laplace distribution. If we want to use acceptance-rejection sampling to use h to sample from f, find c and g in the decomposition $f(t) = cg(t)h(t)$. What is the probability of acceptance?

$$c\, g(t) = \frac{f(t)}{h(t)} = \frac{e^{-t}}{(1+e^{-t})^2} \frac{2}{e^{-|t|}}$$

break problem into 2 pieces, $t \le 0$, $t \ge 0$

$c \ge 1$

$g(t) \in [0,1]$

$P(\text{accept}) = 1/c$